

Vu Truong

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PROFESSIONAL SUMMARY

Hands-on mechanical engineer specializing in autonomous UAS and robotic systems. Experienced across the full design, build, and test lifecycle of competition-grade and field-deployed autonomous aircraft, spanning airframe design, avionics integration, autonomy, and payload systems. Proficient in SolidWorks, Fusion 360, ROS 2, PX4, ArduPilot, Gazebo, and Linux-based systems for mechanical design, autonomous flight, simulation, and hardware integration. Proven team leader driving multidisciplinary projects from concept through flight testing. FAA Part 107 certification in progress.

EXPERIENCE

Maryland Unmanned Aerial System (MUAS)

College Park, MD

President / Team Captain

August 2024 – September 2026

- Led a 10+ member multidisciplinary team (airframe, avionics, autonomy, payload) to design, build, and integrate a competition-ready autonomous sUAS, owning system-level integration and milestone planning.
- Drove iterative mechanical design of airframe structures, component mounts, and payload assemblies in SolidWorks and Fusion 360, then led hands-on fabrication, assembly, and fit-up across the aircraft.
- Performed power budgeting and link-margin analysis across 4 mission profiles to drive battery, motor, and propulsion sizing decisions.
- Led root-cause troubleshooting across mechanical, electrical, and firmware layers through 5+ field test campaigns, iterating hardware against real flight performance.
- Integrated an ArduPilot and ROS2 autonomy and perception stack, coordinating the mechanical and electrical interfaces for onboard sensors and mission payload.

xFoundry Crossfire – Wildfire Response

College Park, MD

Intern, Suppression Autonomy (Drone & Payload)

August 2025 – May 2026

- Modified and integrated a Freefly Systems Alta X airframe for autonomous wildfire suppression, adapting payload-mounting hardware, weight-and-balance, and mechanical drop interfaces to meet mission-specific precision requirements.
- Conducted hardware-in-the-loop testing and flight controller tuning, validating autonomous waypoint following and geofencing logic prior to field deployment.
- Utilized an NVIDIA Isaac Sim and PX4 SITL environment to validate full-system behavior before flight, reducing reliance on physical testing and accelerating development.
- Developed the autonomous flight and perception pipeline (ROS2, Python, PX4) integrating wildfire detection, GPS-corrected navigation, and precision payload-drop control across 3 mission phases.

PROJECTS

Senior Capstone – Autonomous Snow Removal Robot

College Park, MD

Team Member

August 2025 – December 2025

- Designed, built, and tested an autonomous snow removal robot with a 5-person team, taking it from concept to functional prototype in one semester while managing scope and schedule across the full engineering lifecycle.
- Designed and built the drivetrain and actuator subsystems, including mechanical drive layout, motor and component selection, and wiring, integrating motor controllers and power electronics for autonomous operation on uneven terrain.
- Owned component selection and electromechanical subsystem validation, leading integration troubleshooting across mechanical, electrical, and software layers.

BOZOPAD – Custom Macropad | Mechanical Enclosure, PCB Design, Hardware

September 2021 – June 2022

- Designed a custom through-hole PCB, QMK firmware, and 3D-printed housing for a 3×4 macropad, owning the full product lifecycle from schematic capture and mechanical enclosure design through assembly.
- Applied DFM principles to the board layout and enclosure to reduce assembly errors, and authored assembly instructions, packaging, and safety training for repeatable end-user builds.

TECHNICAL SKILLS

CAD & Mechanical Design: SolidWorks, Fusion 360, mechanical drawings & documentation, DFM, 3D printing, PCB design (KiCad)

Hardware & Test: Rapid prototyping, airframe & subsystem integration, hardware-in-the-loop testing, motor controllers, ESCs, power electronics, sensor integration, soldering & rework

Software & Simulation: Python, C++, MATLAB, ROS/ROS2, PX4, ArduPilot, Gazebo, NVIDIA Isaac Sim, Ubuntu/Linux

Tools & Platforms: Git/GitHub, Asana, Microsoft Office, Google G Suite

Certifications: FAA Part 107 (in progress)

EDUCATION

University of Maryland

College Park, MD

B.S., Mechanical Engineering | GPA: 3.52